Week 11

Introduction

As electric vehicles (EVs) transition from niche to mainstream, the seemingly simple act of plugging in a charger belies a sophisticated, intricate digital conversation happening behind the scenes. This week, we delve into the complex world of DC fast charging, specifically focusing on the Combined Charging System 2 (CCS2) standard and the Indian Bharat Electric Vehicle Charging (BEVC) DC001 standard. Understanding these protocols is not just about knowing how to transfer power; it's about appreciating the meticulous orchestration of communication, safety checks, and precise control that ensures a safe, efficient, and interoperable charging experience.

The high power levels involved in DC charging necessitate continuous, reliable information exchange between the EV and the charging station (EVSE). This high-level communication (HLC) is the backbone of modern EV charging, ensuring that both vehicle and infrastructure are synchronized, compatible, and operating within safe parameters. From the moment a charger is connected to the final disconnection, a complex sequence of events unfolds, governed by international and national regulations. Grasping this sequence, the underlying communication protocols like Power Line Communication (PLC) and Controller Area Network (CAN), and the structured approach of the OSI model, is fundamental for anyone involved in the design, operation, or maintenance of EV charging infrastructure, paving the way for a sustainable and interconnected electric mobility future.

Core Concepts

Term	Definition	Significance
`HLC`	High-Level Communication, the advanced digital data exchange between an EV and EVSE, typically over the Control Pilot (CP) and Protective Earth (PE) lines.	Essential for managing complex charging parameters, authentication, and safety checks that cannot be handled by simple PWM signaling.
`OSI Model`	Open Systems Interconnection model, a conceptual framework that standardizes the functions of a telecommunication or	Provides a structured approach to understanding and designing complex communication protocols,

Term	Definition	Significance
	computing system into seven distinct layers.	ensuring interoperability and reliability in CCS2.
`PLC Communication`	Power Line Communication, a method where data is transmitted over existing power lines.	Enables high-level information exchange between EV and EVSE over the Control Pilot (CP) line, crucial for DC charging in CCS2.
`SLAC`	Signal Level Attenuation Characterization, a protocol to establish a logical connection between physically connected PLC modems.	Crucial for preventing crosstalk and ensuring that an EV communicates only with its physically connected EVSE in multi-charger environments.
`CAN Communication`	Controlled Area Network, a message-based, two-wire serial bus communication protocol used for data exchange between electronic control units.	It is the dedicated digital communication method between the EV and EVSE in BEVC-DC001, enabling efficient and robust data sharing.
`Pre-charging`	An initial phase where the EVSE gradually raises its output voltage to match the EV battery voltage.	Prevents large inrush currents and protects the battery and charging system from damage.
`EVSE`	Electric Vehicle Supply Equipment, commonly known as the charging station.	The infrastructure component that supplies electrical energy to charge the EV, managing power delivery and communication.
`BEVC-DC001`	The Bharat Electric Vehicle Charging standard DC001, introduced by the Department	An initial and still prevalent DC fast charging standard in India, primarily

Term	Definition	Significance
	of Heavy Industries, Government of India in 2017.	supporting low-voltage powertrain EVs.
`EV as Master`	In CCS2 protocol messaging, the Electric Vehicle (EV) always initiates requests, acting as the master, while the EVSE responds as the slave.	Simplifies the communication flow and ensures a clear hierarchy in message exchange, preventing conflicts.

Detailed Analysis

1. The CCS2 DC Charging Sequence: A Step-by-Step Orchestration

The **DC Charging (CCS2)-II** lesson outlines the meticulous sequence of events during a CCS2 DC charging session, governed by standards like IEC 61851-23 and AIS 138.

- **Initial State and Connection (Mating)**:
- The process begins with the charger plug unmated, CP line at +12V (State A).
- Upon insertion (t0), the Proximity Pilot (PP) pin detects the connector, transitioning to State B1.
- The vehicle's ECU immobilizes the vehicle, disconnecting the high-voltage battery for safety.
- **Initialization and Handshaking (t0 - t1)**:
- The EVSE activates Pulse Width Modulation (PWM) on the CP line (3-7% duty ratio), moving to State B2 (+9V to -12V pulses).
- High-Level Communication (HLC) begins, superimposing high-frequency signals (PLC) on the 1 kHz PWM.
- The EV (master) sends its maximum DC voltage and current requirements.
- Safety checks monitor DC voltage and compatibility; incompatibility leads to shutdown.
- The EV locks the connector, and the EVSE responds with its capabilities.
- **Cable and Isolation Check (t1 - t4)**:
- The EV signals "ready" by changing its CP state to C2 (+6V) or D (+3V) (t2).
- The EV requests a cable and insulation check.

- The DC EVSE performs an internal isolation check (>100 kΩ) and reports its status.
- Upon successful check, the DC supply indicates "Ready" (t4).
- **Pre-charging Stage (t5 - t8)**:
- The EV requests pre-charge (t5), limiting current to 2A and specifying requested DC voltage.
- The DC supply adapts its output to match (t6).
- The EV's internal disconnecting device closes only if voltage deviation is <20V (t7), preventing inrush current.
- The EV requests full power delivery (t8).
- **Energy Transfer (Main Charging) (t8 - t10)**:
- The EV sets the DC current request (t8).
- The DC supply adapts its output (t8-t9), and current reaches the requested value (t9).
- Continuous monitoring of lock, isolation, voltage, current, and temperature occurs.
- This phase continues until the desired State of Charge (SOC) is reached (t10).
- **Power Down and Disconnection (t10 - t17)**:
- The EV reduces current request (t10-t11), and the DC supply follows, reducing current to <1A.
- The EV requests the DC supply to disable output (t11).
- The EV opens its disconnecting device (t11).
- The DC supply disables output and opens contactors (t12), reporting "Not Ready" (t13).
- The EV changes CP state back to B (t14) and starts isolation monitoring (t15).
- The EV unlocks the connector (t16) only if DC output voltage is <60V, sending a "Session Stop Request."
- The charger plug is removed (t17), returning CP to State A, and PLC communication terminates.

2. Establishing Logical Connections: The SLAC Procedure in CCS2

The **DC Charging (CCS2)-III** lesson focuses on the critical communication protocols, particularly the Signal Level Attenuation Characterization (SLAC)

procedure, which ensures a correct logical connection in multi-charger environments.

- **PLC-based Communication**:
- CCS2 HLC uses Power Line Communication (PLC), specifically HomePlug Green PHY (HPGP), over the CP and PE lines.
- HPGP operates in the 2-28 MHz range, achieving 4.5-10 Mbps data rates.
- Crosstalk is a significant challenge in multi-EVSE stations, where signals can interfere, leading to incorrect logical connections.
- **SLAC Procedure**: This procedure mitigates crosstalk by measuring signal attenuation.
- **Parameter Exchange/Setup**: EV sends `CM_SLAC_PARM.REQ`, EVSEs respond with `CM_SLAC_PARM.CNF` including sounding signal parameters.
- **Discovery of PLC Nodes**: EV broadcasts `CM_START_ATTEN_CHAR.IND` and then sends 10 "sounding messages" (`CM_MNBC_SOUND.IND`). EVSEs measure attenuation for 58 frequency groups and send `CM_ATTN_CHAR.IND` with average values.
- **Matching Decision**: EV analyzes attenuation profiles, choosing the EVSE with the least attenuation. If ambiguous, validation is needed.
- **Validation (Optional)**: EV sends `CM_VALID_CPLT.REQ`. EV performs a "BCB toggle" (rapidly closing/re-opening S2 switch, changing CP voltage between B2 and C2). The physically connected EVSE confirms the number of detected toggles, validating the connection.
- **Key Exchange**: EV sends `CM_SLAC_MATCH.REQ` to the chosen EVSE. EVSE confirms with `CM_SLAC_MATCH.CNF` containing Network Membership Key (NMK) and Network ID (NID). EV sets these keys, and PLC modems reboot, establishing a logical connection.
- **Stateless Address Autoconfiguration (SLAAC)**: After SLAC, SLAAC (an IPv6 mechanism) assigns local-link IP addresses to the EV and EVSE PLC modems, enabling network-layer communication.

3. The OSI Model in CCS2 HLC: A Layered Approach

The **DC Charging (CCS2)-IV** lesson details how the 7-layer Open Systems Interconnection (OSI) model structures the high-level communication in CCS2, as defined by ISO 15118.

- **Physical Layer (Layer 1)**:
- **Function**: Establishes and stabilizes the physical link on the CP and PE wires using HomePlug Green PHY PLC modems.

- **Outcome**: A stable physical link for bidirectional data transfer within the 2-28 MHz frequency band.
- **Data Link Layer (Layer 2)**:
- **Function**: Configures PLC nodes, performs the SLAC procedure to establish a logical connection between physically connected modems, and handles MAC addressing.
- **Outcome**: Error-free transfer of data frames between nodes, with IP packets augmented by MAC addresses.
- **Network Layer (Layer 3)**:
- **Function**: Implements IPv6 for logical addressing (IP addresses) using SLAAC, and ICMPv6 for error messages. Handles routing of data packets.
- **Outcome**: All communicating nodes obtain unique IP addresses, enabling efficient routing across the network.

4. Completing the OSI Model and CCS2 Protocol Messaging

The **DC Charging (CCS2)-V** lesson completes the OSI model discussion and dives into the detailed messaging sequence of CCS2.

- **Transport Layer (Layer 4)**:
- **Function**: Ensures reliable (TCP), fast (UDP), and secure (TLS) communication between nodes.
- **Outcome**: A reliable, fast, and safe data connection between all communicating nodes.
- **Session Layer (Layer 5)**:
- **Function**: Implements the Vehicle-to-Grid Transport Protocol (V2GTP) to establish, manage, and terminate communication sessions.
- **Outcome**: A stable charging session connection between the EV and EVSE.
- **Presentation Layer (Layer 6)**:
- **Function**: Handles data encoding/decoding using Efficient XML Interchange (EXI) format, ensuring message validity and compatibility.
- **Outcome**: Correct data exchange between EV and EVSE.
- **Application Layer (Layer 7)**:
- **Function**: Manages all messages for the charging process (identification, pre-charge, charge, security checks), selects charging standards (e.g., DIN SPEC 70121 or ISO 15118). EV sends `supportedAppProtocolReq`, EVSE responds with `supportedAppProtocolRes`.
- **Outcome**: Initialization and configuration of the charge process.

- **SECC Discovery Protocol (SDP)**: After SLAC, the EV uses UDP-based SDP to broadcast an `SDP Request` and obtain the SECC's IP address and TCP port from the `SDP Response`.
- **CCS2 Protocol Messaging**: The EV acts as the master, initiating requests, and the EVSE responds.
- **Communication Setup**: Application protocol selection, session setup (`SessionSetupReq` with `sessionID = 0`).
- **Identification, Authentication, Authorization**: Service discovery (`ServiceDiscoveryReq`), service details (`ServiceDetailReq`), payment service selection, and authorization (`AuthorizationReq`).
- **Target Setting and Charge Scheduling**: `ChargeParameterDiscoveryReq` to exchange max voltage, current, power. `EVSE processing` field indicates `ongoing` or `finished`.
- **Charging Loop / Re-scheduling**: `CableCheckReq` (insulation resistance), `PreChargeReq` (voltage matching <20V), `PowerDeliveryReq` (EV closes contactor), `CurrentDemandReq` (continuous adjustment).
- **End of Charging**: EV sends `PowerDelivery` (end), optional `WeldingDetectionReq`, then `SessionStopReq`.

5. BEVC-DC001: India's DC Charging Standard

The **DC Charging System (BEVC-DC001)-I** lesson introduces India's specific DC charging standard for low-voltage EVs, highlighting its unique communication approach.

- **Context and Specifications**:
- Introduced in 2017 for two-wheelers, three-wheelers, and early four-wheelers.
- Input: 415V AC (three-phase, five-wire).
- Output: Max 10 kW for 48V systems (200A), 15 kW for 72V systems (200A).
- Connector: GB/T 20234.3 with dedicated DC+, DC-, Charging Confirmation (CC1, CC2), CAN (S+, S-), Auxiliary (A+, A-), and Protective Earth (PE) pins.
- EVSE Types: Type 1 (10 kW/3.3 kW) and Type 2 (10 kW/15 kW for Output 1, 3.3 kW for Output 2).
- Functional Requirements: Voltage accuracy ±0.5%, current accuracy ±1A (≤50A), overvoltage protection, >92% efficiency, >0.9 power factor, 200A max current.
- **CAN Communication**:

- Unlike CCS2's PLC, BEVC-DC001 uses a dedicated Controller Area Network (CAN) for digital communication.
- Uses CAN high (CAN_H) and CAN low (CAN_L) lines (S+, S- pins) with shielded twisted pair and 120-ohm termination.
- Physical connection detection and locking are hardware-based via CC1, CC2 pins.
- Layered Model:
- **Physical Layer**: Derived from SAEJ1939-11/ISO 11898-1 (termination, voltage, speed).
- **Data Link Layer**: Derived from SAEJ1939-21 (addressing, CAN data frame formats).
- **Application Layer**: From GB/T 27930-2015 (parameter representation, PDU formats, customized for India).
- Speed: 250 kbps, CAN 2.0b protocol (29-bit identifier).
- Participants: Charger controller (address 86 decimal) and vehicle's BMS (address 244 decimal).
- This CAN channel is separate from in-vehicle CAN systems.
- **Benefits of CAN**:
- **Low Cost**: Reduces wiring, weight, and manufacturing costs.
- **Centralized Communication**: Enables central diagnosis and configuration.
- **Robustness**: Resistant to EMI due to differential signaling and twisted pair.
- **Efficiency**: Priority-based message IDs ensure critical data transmission.
- **Error Capability**: CRC for error checking; faulty messages are discarded.
- **Flexibility**: Easy to add/remove nodes with minimal software changes.
- **Speed**: Up to 1 Mbps (40m) or 125 kbps (100m).
- **Multi-Master Configuration**: Any node can initiate communication if the bus is free, with dynamic master role based on message priority.

Key Takeaways

- [ ] DC CCS2 charging involves a sophisticated, multi-stage sequence from mating to unmating, governed by HLC over the CP line.
- [ ] The Proximity Pilot (PP) pin immobilizes the vehicle upon connection, and pre-charging prevents damaging inrush currents by matching voltages.
- [ ] High-level communication in CCS2 relies on Power Line Communication (PLC) using HomePlug Green PHY over the CP and PE lines.

- [ ] The SLAC (Signal Level Attenuation Characterization) procedure is crucial in CCS2 to logically connect the correct EV and EVSE PLC modems, especially in multi-charger environments, by measuring signal attenuation and using BCB toggles for validation.
- [ ] The CCS2 HLC adheres to the 7-layer OSI model, with ISO 15118 defining protocols for each layer, from physical connection to application-level messaging.
- [ ] In CCS2, the EV acts as the communication master, initiating all requests, while the EVSE responds.
- [ ] BEVC-DC001 is India's DC charging standard for low-voltage EVs, utilizing a GB/T 20234.3 connector and dedicated CAN communication.
- [ ] BEVC-DC001's CAN communication is derived from SAEJ1939 and GB/T 27930, operating at 250 kbps with fixed addresses for the charger and BMS.
- [ ] CAN communication offers significant benefits including robustness, efficiency, error checking, and reduced wiring complexity compared to point-to-point wiring.

Conclusion

This week's exploration into the intricacies of DC charging systems, particularly CCS2 and BEVC-DC001, reveals that modern EV charging is far more than just plugging in. It's a complex, intelligent grid of communication protocols, safety mechanisms, and power management strategies. The detailed sequences, from the initial handshaking and compatibility verification in CCS2 to the robust CAN communication in BEVC-DC001, underscore the engineering marvels that ensure safe, reliable, and efficient energy transfer.

Understanding these foundational communication layers and protocols, whether it's the OSI model's structured approach in CCS2 or the dedicated CAN bus in BEVC-DC001, is paramount for anyone involved in the EV ecosystem. This knowledge not only facilitates the development and deployment of compliant charging solutions but also lays the groundwork for future innovations. As electric mobility continues its rapid expansion, the reliability and intelligence embedded in these high-level communication standards will be critical enablers for advanced features like smart charging, vehicle-to-grid (V2G) capabilities, and optimized energy management, seamlessly integrating EVs into a sustainable and interconnected energy future.